

West Asia War 2026 — Updated Project Roadmap & README List

Last Updated: 2026-05-12

Current Phase: Phase 2, Step 9 (Data Merging)

Complete Updated Roadmap

Phase 0 — Project Setup  COMPLETE

Phase 1 — Data Ingestion  COMPLETE

Step 4 — NASA FIRMS  Complete

File: `src/kinetic_pulse.py`

Features:

- NRT mode for realtime polling (last 1-10 days)
- Archive mode for historical backfill (VIIRS_SNPP_SP, 10-day chunks)
- Silent failure guard (validates raw response before parsing)

Status: Realtime operational. Historical pending NASA archive download email.

Step 5 — GDELT Sentiment  Complete

Files: `src/diplomatic_sentiment.py`, `notebooks/distribution_analysis.ipynb`,
`notebooks/gpu_utilisation_test.ipynb`

Features:

- Dual signal architecture (DistilBERT + GDELT tone)
- GMM regime weights (hostile 73% / diplomatic 27% during ceasefire)
- Full distribution statistics (percentiles, skew, bloc breakdown, theme breakdown)
- Country×theme confounding fix
- ThreadPoolExecutor parallelization (3 workers, 1s delay for realtime)
- **NEW:** Stealth mode for historical backfill (1 worker, 5s delay with exponential backoff)

Key Findings:

- Normality rejected by Shapiro-Wilk, D'Agostino-Pearson, KS tests
 - Bimodal distribution confirmed — DistilBERT binary architecture artifact
 - GMM fitted — two regimes identified
 - GPU: Sequential loop 143.1 art/sec, Dataset approach 99.5 art/sec → Sequential kept
-

Step 6 — Yahoo Finance Economic Signals Complete

Files: `src/economic_signals.py`, `notebooks/economic_signals_analysis.ipynb`

Key Findings:

- Weekend gap solved via forward-fill (60 trading days → 82 continuous days)
- Stationarity tested with joint ADF + KPSS
- **Brent Crude:** trend-stationary at raw level (deterministic war-driven trend)
- **S&P 500:** permanent structural break confirmed by both tests → **DROPPED FROM MODEL**
- **Gold, USD/ILS, VIX:** stationary after first differencing
- First differences selected over log returns for model inputs
- **VIX designated sole global panic proxy** (forward-looking implied volatility)
- S&P 500 kept as **display-only % drawdown** from pre-war peak (Feb 27 2026)

Signals Entering Model:

- Brent Crude daily change
 - Gold daily change
 - USD/ILS daily change
 - VIX daily change
-

Step 7 — GDELT Kinetic Events Complete

Files: `src/gdelt_kinetic.py`, `notebooks/gdelt_kinetic_analysis.ipynb`

Architecture Decision: ACLED dropped (paywall) → replaced by GDELT Event Database

- **V1 daily files** for historical backfill (100% coverage, no sampling assumption)
- **V2 15-minute files** for realtime polling

CAMEO Codes Ingested: 18 (Assault), 19 (Fight), 20 (Mass Violence)

Critical Findings:

- **GDELT × FIRMS correlation:** $r = -0.202$, $p = 0.089$
 - **Interpretation:** Signals are independent, not redundant
 - GDELT = narrative/reporting layer
 - FIRMS = physical/thermal layer
 - **Validates Dual-Signal Veto System** — cross-validation is essential precisely because signals are orthogonal
 - **Reporting lag:** +6 days (GDELT news trails FIRMS thermal detection at peak CCF)
 - **Correction applied in Step 9:** Subtract 6 days from GDELT event dates before merging
 - **Capital-snap coordinate rate:** 10.2% (low spatial bias)
 - Spatial Anchoring still implemented in Step 9 but bias is manageable
-

Step 8a — Polling Loop Complete

File: `src/data_collector.py`

Master orchestrator calling all 4 modules every 15 minutes.

Modules Called:

- `kinetic_pulse.run_realtime()`
- `gdelt_kinetic.run_realtime()`
- `diplomatic_sentiment.run_realtime()`
- `economic_signals.run_realtime()`

Usage:

```
bash

python src/data_collector.py --mode once      # Single poll cycle
python src/data_collector.py --mode loop      # Continuous every 15 min
```

Step 8b — Snapshot System Complete

File: `src/snapshot_manager.py`

Saves dated model output JSON to `results/snapshots/YYYY-MM-DD.json` every run.
Committed to GitHub. Powers historical replay slider on dashboard (Step 19).

Snapshot Schema:

json

```
{
  "date": "2026-03-19",
  "data_type": "realtime" | "historical_reconstruction",
  "sentiment": {...},
  "kinetic": {...},
  "economic": {...},
  "model_output": {...},          // Phase 3 - null until then
  "confidence_interval": {...}    // Phase 3 - null until then
}
```

Status: 100 snapshots generated (Feb 01 → May 11)

Step 8c — Historical Backfill Complete

File: `src/historical_backfill.py`

Orchestrates historical reconstruction Feb 01 → today.

Run Order:

1. Economic signals (yfinance, single call, ~30 seconds)
2. GDELT kinetic (V1 daily, 1 file/day, ~2 minutes)
3. FIRMS archive (pending NASA email download)
4. Diplomatic sentiment (14-day chunks, ThreadPool, ~10-15 minutes)
5. Generate snapshots (merge CSVs → daily JSONs)

Usage:

bash

```
python src/historical_backfill.py --mode full          # All modules
python src/historical_backfill.py --mode economic     # Individual
python src/historical_backfill.py --mode firms
python src/historical_backfill.py --mode gdelt_kinetic
python src/historical_backfill.py --mode sentiment
python src/historical_backfill.py --mode snapshots
```

Phase 2 — Feature Engineering ← CURRENT PHASE

Step 9 — Data Merging 🖱 ACTIVE

Files: `src/data_merger.py`, `notebooks/data_merging_analysis.ipynb`

Join all 4 sources on `date` into master DataFrame.

Key Operations:

1. GDELT 6-day lag correction

- Subtract 6 days from GDELT event dates before joining
- Example: GDELT reports event on Mar 15 → reassigned to Mar 09 (actual occurrence)

2. Weekend forward-fill

- Markets close on weekends, wars don't
- Economic signals forward-filled from Friday (already applied in Step 6)

3. Missing data flagging

- Days with no FIRMS data → `firms_data_missing = True`
- Days with no GDELT data → `gdelt_data_missing = True`
- Days with no sentiment data → `sentiment_data_missing = True`
- Never dropped — flagged for uncertainty propagation in Phase 3

4. S&P 500 walled off

- SP500 column retained as display-only % drawdown
- Never enters model features — clearly marked

PENDING:

- **BallTree Spatial Anchoring** (Haversine 50km radius, ± 2 day window)
 - Cross-match GDELT coordinates against FIRMS thermal anomalies
 - Overwrite capital-city coordinates with validated lat/lon
- **Audit script** for sentiment data quality checks
- **Duplicate date handling** in sentiment CSV (`drop_duplicates`)

Assumption (to be tested in notebook): Joining on date after 6-day GDELT lag correction produces a clean master DataFrame without information leakage. Forward-fill is sufficient for weekend gaps. Missing days flagged, not dropped.

Step 10 — Normalization

MinMaxScaler to 0-1. Fit on training window only — not full dataset — to avoid leakage.

Assumption to test: All variables can be linearly scaled without distortion from outliers.

Notebook: `notebooks/normalization_analysis.ipynb`

Step 11 — Lagging

Known lags to implement:

- Oil price lags strike events ~3 days
- Diplomatic sentiment lags ceasefire ~7-14 days
- **GDELT lags real-time events 6 days** (confirmed in Step 7)

Assumption to test: These lag lengths are correct for this specific conflict.

Test: Cross-correlation analysis to find empirical optimal lag per variable.

Step 12 — Shock Detection

Dual-Signal Veto System implementation:

Scan FIRMS thermal data for Poisson Jump events. Flag `shock_detected = True` when:

1. **FIRMS thermal spike** — $FRP > 2\sigma$ above 7-day rolling mean, AND
2. **GDELT kinetic event co-occurrence** — within 50km radius + ± 2 day window

Neither signal alone is sufficient. Both must fire to trigger validated Kinetic Shock.

Assumption to test: 2σ threshold is optimal.

Test: Compare detected shock dates against known ACLED strike events. Tune threshold to maximize detection rate while minimizing false positives.

Notebook: `notebooks/shock_detection_analysis.ipynb`

Step 13 — PCA

Reduce 20+ variables to 3-4 principal components.

Assumption to test: Linear relationships between variables (PCA assumes linearity).

Test: Correlation matrix heatmap. If non-linear, consider UMAP instead.

Notebook: `notebooks/pca_analysis.ipynb`

Phase 3 — Core Models

Step 14 — Lanchester ODE Model

File: `src/lanchester_model.py`

Square Law ODEs solved with `scipy.integrate.solve_ivp`.

Probabilistic Casualties: GDELT has no exact casualty counts. Lanchester k-coefficients (lethality parameters) approximated via **Negative Binomial distributions** mapped from GDELT CAMEO codes:

- CAMEO 18 (Assault) → $\text{NegBin}(\mu=5, \sigma=3)$
- CAMEO 19 (Fight) → $\text{NegBin}(\mu=12, \sigma=8)$
- CAMEO 20 (Mass Violence) → $\text{NegBin}(\mu=50, \sigma=30)$

Assumption to test: Square Law applies throughout (assumes modern aimed combat). Linear Law may apply for insurgency phases.

Test: Compare predicted attrition rates against ACLED confirmed casualty data.

Notebook: `notebooks/lanchester_validation.ipynb`

Step 15 — Stochastic Shock Integration

Wrap ODE solver with Poisson Jump logic. Shock events from Step 12 trigger discrete step-downs in force strength and temporary k-coefficient decay.

Step 16 — Diplomatic Switch

Two-trigger "Trump Pause" logic:

- Primary: GMM hostile weight drops below 0.50 for 3+ consecutive days
- Secondary: sentiment_diplomatic crosses above 0.0 for 3+ consecutive days

Both must hold before ceasefire switch activates.

Step 17 — Monte Carlo Engine

File: `src/monte_carlo.py`

10,000 iterations. Each randomly samples:

- Initial force strengths from distribution around OSINT estimates

- k-coefficients from historical variance
- Shock timing from Poisson distribution with rate λ tuned to Feb 28 → Apr 7 conflict data

Assumption to test: 10,000 iterations is sufficient for PDF convergence.

Test: Compare PDFs at 1,000 / 5,000 / 10,000 / 50,000 iterations. Find minimum count where PDF stops changing materially.

Notebook: `notebooks/monte_carlo_analysis.ipynb`

Step 18 — Probability Density Function

Gaussian KDE over Monte Carlo outcomes. Four outcome categories:

- Negotiated ceasefire
 - Iranian capitulation
 - Stalemate / frozen conflict
 - Regional escalation
-

Phase 4 — Dashboard

Step 19 — Dashboard Build

File: `dashboard/app.py` — Plotly Dash

Eight panels:

1. Force attrition curves with confidence bands
2. Sentiment trend (DistilBERT avg + GMM hostile weight + GDELT tone)
3. Economic signals (Brent, VIX, ILS, Gold daily changes — S&P 500 display-only drawdown)
4. NASA FIRMS heat map (geospatial thermal anomalies)
5. Monte Carlo PDF — headline output
6. Sensitivity tornado chart
7. **Historical replay slider** (Feb 28 → today, loads snapshot JSONs)
8. Backtesting panel (forecast vs actual, Brier Score, calibration curve)

Key conflict events marked as vertical lines on all time-series charts.

Step 20 — Sensitivity Analysis

One-at-a-time variable ranking. Tornado chart.

Step 20b — Static Dashboard Export

Single `results/dashboard_export.html` committed to GitHub.

Recruiters open one file — zero setup required.

Phase 5 — Backtesting & Uncertainty

Step 21 — Rolling Window Backtesting

- Train on Feb 28 → Mar 19
 - Predict Mar 19 → Apr 2
 - Compare against ACLED ground truth
 - Roll window forward by 2 weeks, repeat
 - Compute Brier Score per window
 - Build calibration curve
-

Step 22 — Uncertainty Decomposition

Epistemic vs aleatoric uncertainty separated and displayed as distinct bands.

Step 23 — Three-Scenario Analysis

Bear (regional escalation), Base (frozen conflict/stalemate), Bull (rapid resolution).

Step 24 — Model vs Naive Baseline

Compare against "nothing changes" baseline. Must demonstrate statistically significant improvement.

Phase 6 — Agent-Based Modelling

Step 25 — Mesa ABM

File: `src/abm_synthetic.py`

Synthetic data for censored periods and ACLED reporting lags. Tagged explicitly — never contaminates real data.

Phase 7 — Automation & Polish

Step 26 — GitHub Actions

File: `.github/workflows/data_update.yml`

Daily scheduled workflow:

1. Pull fresh data from all four sources
 2. Run model
 3. Generate today's snapshot JSON
 4. Update `latest_output.json`
 5. Regenerate static dashboard HTML
 6. Commit all back to repo
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Step 27 — Final README Overhaul

Last step. Uses everything from README additions list below.

README Additions List (For Step 27)

✅ **COMPLETED — Ready to Document**

Statistical Analysis

- **Distribution analysis plots** (histogram, Q-Q, GMM fit)
- **GPU benchmark table**
 - RTX 3050 4GB: Sequential loop 143.1 art/sec vs Dataset approach 99.5 art/sec
 - Sequential loop kept — Dataset overhead outweighs benefits on laptop GPU
- **ADF + KPSS stationarity test results table** (joint verdicts)
- **Brent Crude trend-stationarity finding**
 - Trend-stationary at raw level (deterministic war-driven trend)

- No unit root — price level is stationary around a deterministic trend
- **S&P 500 contradictory log returns finding**
 - ADF power loss under structural break
 - Both raw level and log returns fail stationarity
- **S&P 500 dropped from model**
 - Permanent structural break confirmed by both ADF and KPSS
 - Kept as display-only % drawdown from pre-war peak (Feb 27 2026)
- **VIX designated as forward-looking leading indicator**
 - Sole global panic proxy
 - Forward-looking implied volatility measure
- **S&P 500 display-only drawdown visualization** (pre-war peak = Feb 27 2026)

Architecture & Methodology

- **Dual-Signal Veto System**
 - FIRMS provides physical proof (thermal spikes)
 - GDELT provides narrative (who fired at whom)
 - Neither can trigger Kinetic Shock alone — cross-validation required
 - Industrial accidents (FIRMS noise) and political rhetoric (GDELT noise) filtered out
- **ACLED → GDELT pipeline transition note**
 - ACLED deprecated due to enterprise paywall
 - GDELT V1 daily for historical (one complete file/day, 100% coverage)
 - GDELT V2 15-min for realtime (updates every 15 minutes)
- **Spatial Anchoring methodology**
 - BallTree with Haversine metric (50km radius, ± 2 day window)
 - GDELT capital-city coordinates overwritten with FIRMS-validated lat/lon
 - Cross-validation only when both signals co-occur spatiotemporally
- **Probabilistic Casualties (CAMEO → NegBin)**
 - GDELT lacks exact fatality integers
 - Lanchester k-coefficients simulated via Negative Binomial distributions
 - CAMEO 18 (Assault) → NegBin($\mu=5, \sigma=3$)
 - CAMEO 19 (Fight) → NegBin($\mu=12, \sigma=8$)
 - CAMEO 20 (Mass Violence) → NegBin($\mu=50, \sigma=30$)

Key Findings

- **GDELT × FIRMS independence finding**
 - Pearson $r = -0.202$, $p = 0.089$ (weak negative, not statistically significant)
 - **Interpretation:** Signals are genuinely independent, not redundant
 - GDELT captures narrative/reporting layer
 - FIRMS captures physical/thermal layer
 - **Validates Dual-Signal Veto** — independence proves cross-validation is essential
 - Each signal captures what the other misses
 - **GDELT 6-day reporting lag**
 - Peak cross-correlation function (CCF) at lag = +6 days
 - GDELT news reporting trails FIRMS thermal detection by 6 days
 - Correction: subtract 6 days from GDELT event dates before merging (Step 9)
 - **Capital-snap coordinate rate: 10.2%**
 - Low spatial bias in GDELT coordinates
 - GDELT defaults unknown locations to capital city ~10% of the time
 - Spatial Anchoring still implemented but bias is manageable
-

PENDING — Not Yet Implemented

Documentation

- **Conflict timeline table** (ground truth validation markers)
 - Feb 28: Operation Epic Fury — War begins, Khamenei assassinated
 - Mar 01: Iran confirms Khamenei death, Mojtaba named successor
 - Mar 17: Limited ground invasion begins
 - Mar 31: Market-implied ceasefire signal — S&P 500 structural break
 - Apr 07: US-Iran ceasefire declared
 - Apr 08: IDF blitz on Lebanon strains ceasefire
 - Apr 13: US counter-blockade of Strait of Hormuz
- **Data architecture explanation** (4-layer system diagram)
 1. Kinetic Layer (FIRMS + GDELT Event)
 2. Economic Layer (Brent, VIX, Gold, USD/ILS)
 3. Socio-Political Layer (GDELT GKG + DistilBERT + GMM)
 4. Hybrid Layer (Mesa ABM for synthetic data)

- **Historical reconstruction note**
 - 100 snapshots generated (Feb 01 → May 11)
 - Saved to `results/snapshots/YYYY-MM-DD.json`
 - Powers dashboard historical replay slider

Model Evaluation (Phase 5)

- **Brier Score, calibration curve, rolling window table**
- **Sensitivity tornado chart**
- **Monte Carlo PDF screenshot**

Deployment (Phase 7)

- **Dashboard screenshot / GIF**
 - **Static HTML export link**
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Current Folder Structure

```
D:\Projects\west-asia-war-2026\  
├─ data\                                ← gitignored  
│   ├── firms_raw.csv                  ✓ (awaiting NASA email for full Feb-May)  
│   ├── gdelt_kinetic_raw.csv          ✓  
│   ├── gdelt_sentiment_daily.csv     ✓  
│   ├── economic_raw.csv              ✓  
│   ├── sentiment_realtime.csv        ✓  
│   ├── economic_realtime.csv         ✓  
│   ├── firms_realtime.csv            ✓  
│   └─ master_df.csv                  ← Step 9 output (pending)  
├─ notebooks\  
│   ├── distribution_analysis.ipynb    ✓  
│   ├── gpu_utilisation_test.ipynb    ✓  
│   ├── economic_signals_analysis.ipynb ✓  
│   ├── gdelt_kinetic_analysis.ipynb  ✓  
│   └─ data_merging_analysis.ipynb    ← Step 9 (pending)  
├─ src\  
│   ├── __init__.py  
│   ├── kinetic_pulse.py              ✓  
│   ├── diplomatic_sentiment.py       ✓  
│   └─ economic_signals.py           ✓
```

— gdel_t_kinetic.py	✓
— data_collector.py	✓ Step 8a
— snapshot_manager.py	✓ Step 8b
— historical_backfill.py	✓ Step 8c
— data_merger.py	✎ Step 9 (active)
— lanchester_model.py	← Step 14 (next)
— results\	
— snapshots\	✓ 100 JSONs (Feb 01 → May 11)
— latest_output.json	✓
— dashboard\	
— app.py	← Step 19
— tests\	
— .env	
— .gitignore	
— requirements.txt	
— README.md	

Research Methodology — Applied To Every Step

1. **State assumption explicitly** before coding
2. **Build naive version first**
3. **Question the assumption**
4. **Test formally in notebook**
5. **Document finding in markdown cell**
6. **Update production code** based on finding
7. **Commit with meaningful message** linking both

Next Immediate Actions

1. **Answer Gemini handoff questions:**
 - Did diplomatic sentiment historical backfill actually run overnight?
 - Does current `diplomatic_sentiment.py` have exponential backoff?
 - Check for duplicate dates in `gdel_t_sentiment_daily.csv`
 - Confirm if BallTree spatial anchoring was implemented or just planned
2. **Complete Step 9:**
 - Add `drop_duplicates` to `_load_sentiment()` in `data_merger.py`
 - Implement BallTree spatial anchoring (if not already done)

- Run `python src/data_merger.py` to generate `master_df.csv`
- Build `notebooks/data_merging_analysis.ipynb` to validate merge

3. Phase 3 — Begin Lanchester ODEs

- `src/lanchester_model.py`
- Probabilistic Casualties via NegBin distributions
- Shock detection integration

Next Commit Message Template

```
bash
```

```
git add .  
git commit -m "feat: Phase 1 complete — Steps 8a+8b+8c done. 100 historical snapshot  
git push origin main
```